

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2022 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

4049/02

Paper 2

12 September 2022

2 hour 15 minutes

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Calculator Model

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Total Marks

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}ab \sin C$$

1 The equation of a curve is $xy = mx + 4$, where m is a constant.

(a) Find the set of values of m for which the curve and the line $x + y = 2m$ do not intersect. [4]

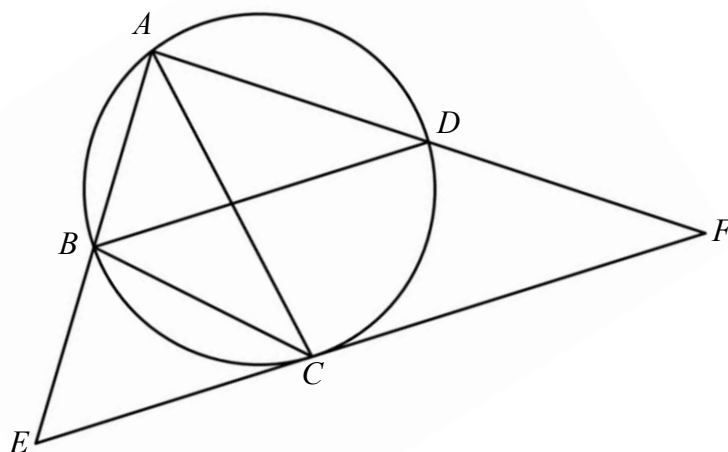
(b) Show that, for all values of m , the line $y = x + 1$ intersects the curve at two distinct points. [2]

- 2 (a) Write down, and simplify, the first 3 terms in the expansion of $\left(1 - \frac{x}{2}\right)^8$ in ascending powers of x . [2]

The expansion of $(1+x)^n \left(1 - \frac{x}{2}\right)^8$, where n is a positive integer greater than 3, has no term in x^2 .

- (b) Find the coefficient of x in the expansion of $(1+x)^n \left(1 - \frac{x}{2}\right)^8$. [5]

3



The diagram shows a circle passing through the points A , B and D . The point E lies on AB produced such that $AB = BE$. The point C lies on the circle and EC is a tangent to the circle. The tangent EC meets AD produced at F and $AD = DF$. Prove that AC bisects angle EAF . [5]

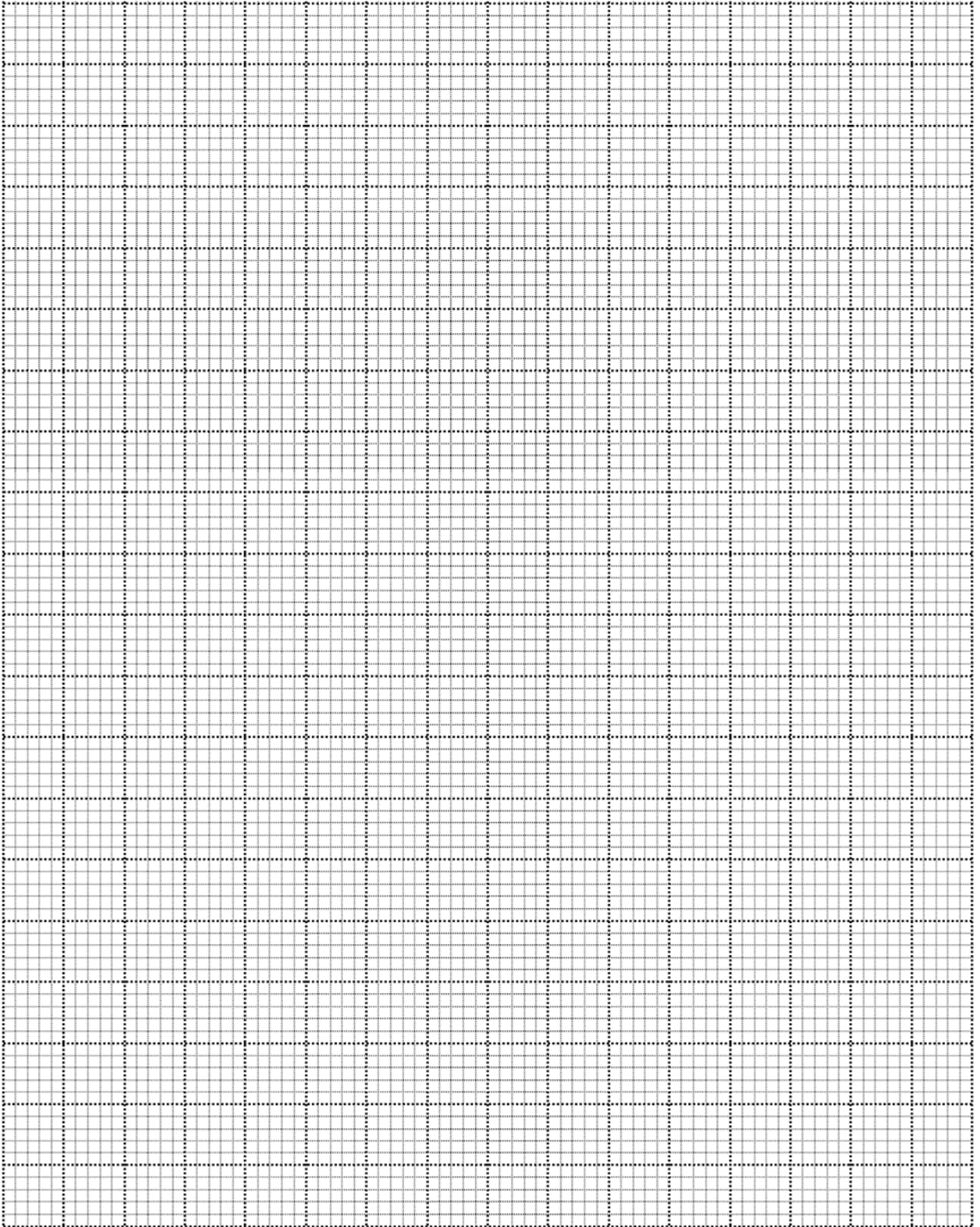
- 4 (a) A viral outbreak occurs in a small town. The number of infected people, N , in the small town after t days can be modelled by the formula $N = ab^t$, where a and b are constants. Values of N for different values of t have been collected. Explain how a straight line graph can be drawn to represent the formula. State the axes of the graph and how the values of a and b can be obtained from the line. [4]

- (b) In an experiment, a student recorded the time taken, t seconds, for an object to travel a distance of d metres. The table shows the recorded values of d and t .

d (metres)	2.00	4.00	6.00	8.00
t (seconds)	1.30	1.95	2.45	2.87

The student believed that d and t can be modelled by an equation of the form $d = pt + qt^2$, where p and q are constants.

- (i) On the grid on page 7, draw a suitable straight line graph. [3]
- (ii) Use the graph to estimate the value of p and of q . [3]



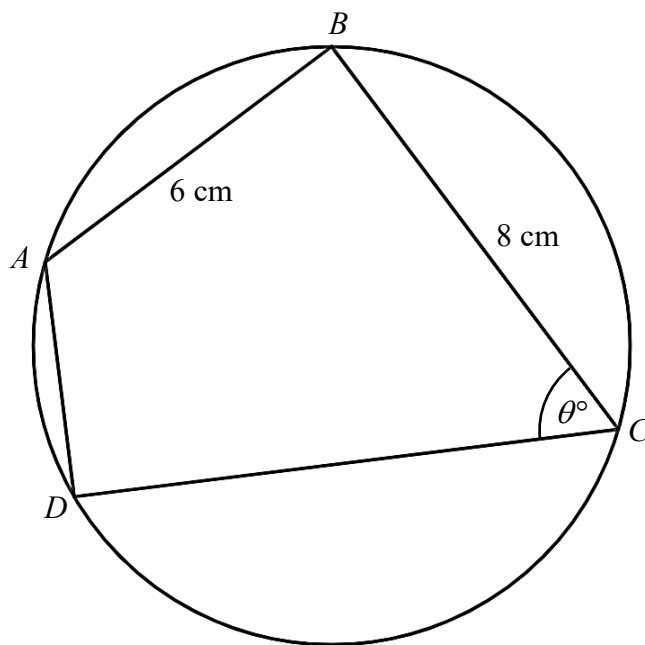
5 The equation of a polynomial is given by $p(x) = 3x^3 + 2x^2 - 7x + 2$.

(a) Find the remainder when $p(x)$ is divided by $2x + 1$. [2]

(b) Solve the equation $p(x) = 0$. [4]

- (c) Use your answer to **part (b)** to solve the equation $2(4^{3y}) - 7(4^{2y}) + 2(4^y) + 3 = 0$. [4]

6



The diagram shows the design of a logo made up of a circle and a quadrilateral $ABCD$.

A , B and C are fixed points such that $AB = 6$ cm and $BC = 8$ cm.

AC is a diameter of the circle. BC makes an angle θ° with the line CD . The angle θ° can vary in such a way that the point D lies on the circle between points A and C .

- (a) Show that the perimeter of $ABCD$, P cm, is given by $P = 14 + 14 \sin \theta + 2 \cos \theta$. [4]

- (b) Express P in the form of $14 + R \sin(\theta + \alpha)$, where $R > 0$ and α is an acute angle. [4]

- (c) State the maximum value of P and find the corresponding value of θ . [3]

- 7 The equation of the tangent T to a circle at the point $(6, -3)$ is $3y + 2x = 3$. The line with equation $4y + x + 20 = 0$ is a normal to the circle.

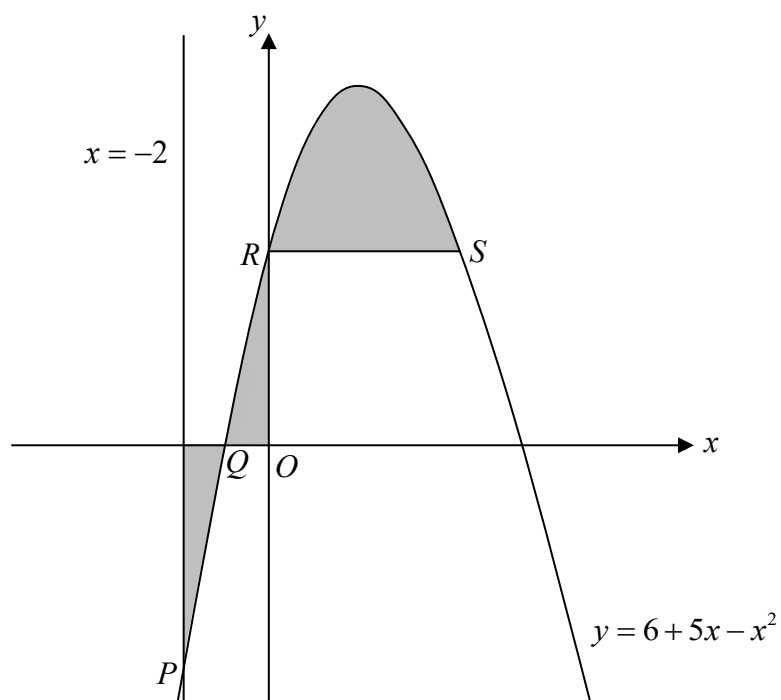
(a) Showing all working, find the equation of the circle.

[7]

P is the point on the circle that is nearest to the y -axis.

- (b) Find the coordinates of the point at which the tangent at P and the tangent T intersect. [2]

8



The diagram shows part of the curve of $y = 6 + 5x - x^2$. The line $x = -2$ intersects the curve at the point P . The curve meets the x -axis and y -axis at the points Q and R respectively. The point S lies on the curve such that the line RS is parallel to the x -axis.

Find

(a) the coordinates of Q and S ,

[3]

(b) the total area of the shaded regions.

[6]

- 9 A particle, moving in a straight line, passes through a fixed point O . The velocity, v m/s, of the particle t seconds after passing through O , is given by $v = 20e^{-0.2t} - 2$.

The particle comes to instantaneous rest at the point P .

Find

- (a) the value of t when the particle reaches P , [3]

- (b) the acceleration of the particle at P , [3]

- (c) the distance travelled by the particle in the first 16 seconds.

[6]

10 (a) Differentiate $\ln(\sin x)$ with respect to x .

[2]

(b) Show that $\frac{d}{dx}(x \cot x) = \cot x - x \operatorname{cosec}^2 x$.

[3]

(c) Using the results from **parts (a) and (b)**, show that

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} x \operatorname{cosec}^2 x \, dx = \frac{(2\sqrt{3}-3)\pi}{12} + \frac{1}{2} \ln 2. \quad [6]$$

END OF PAPER

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